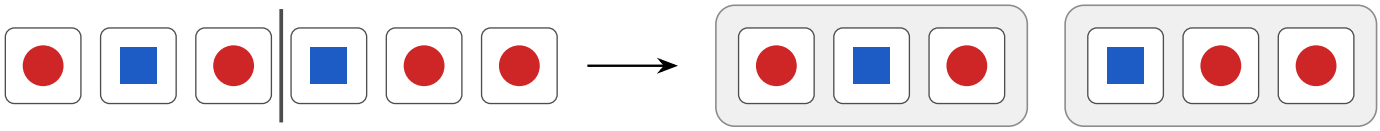
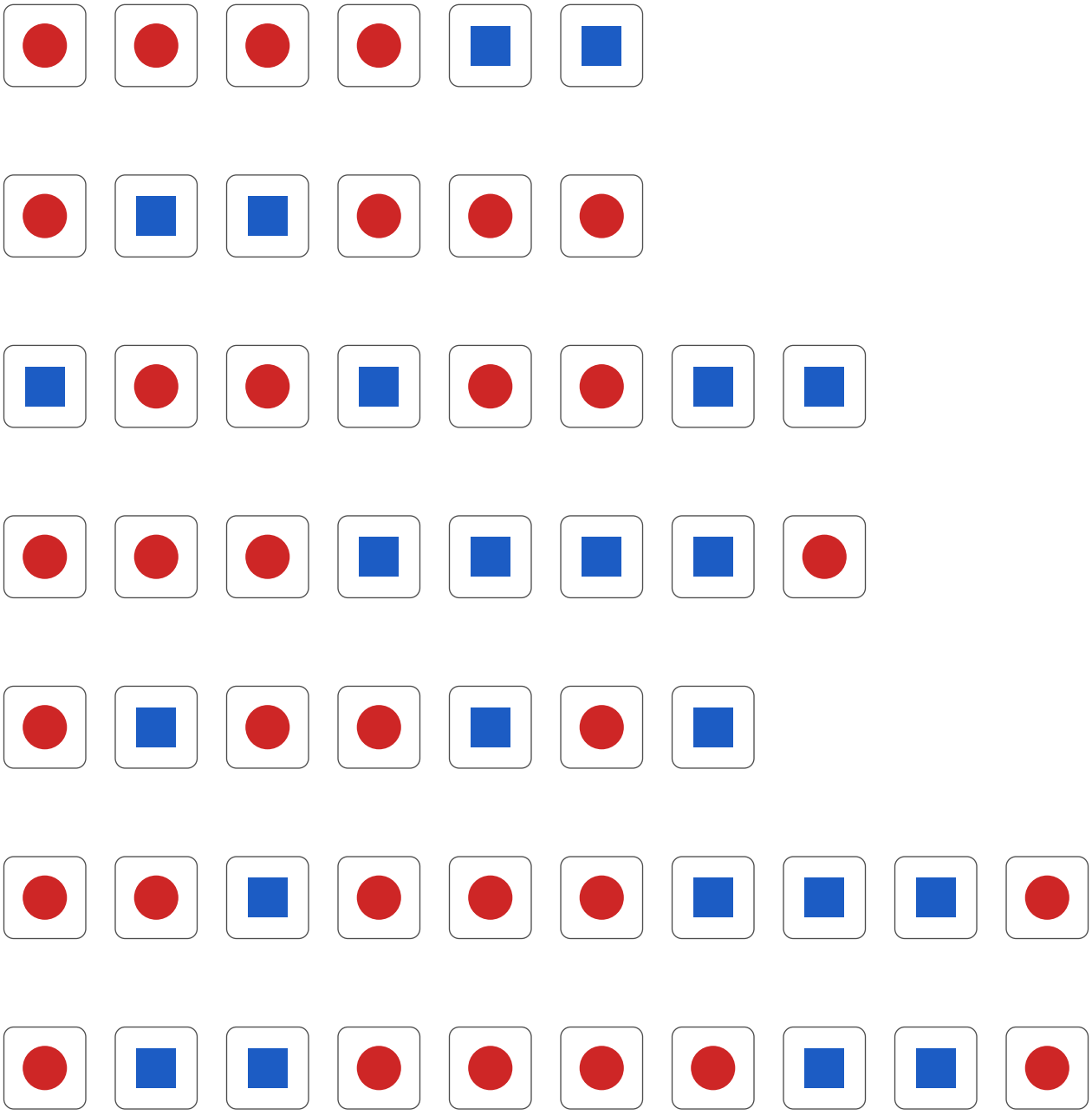
A train is a row of goods cards. To share a train, cut it at some of the gaps between neighbouring cards, so that every card stays whole, and put each piece into one of two trays. A tray may get more than one piece. Both trays must get the same number of red cards and the same number of blue cards.

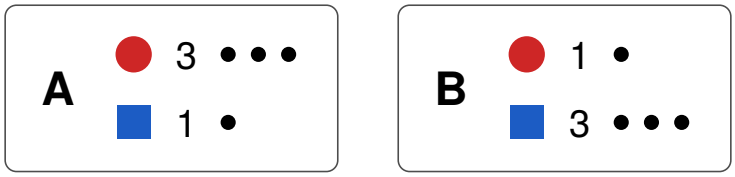


Problem 1: Find every train you can make with two red cards and two blue cards, counting only the order of the colours. Share each one with as few cuts as you can.

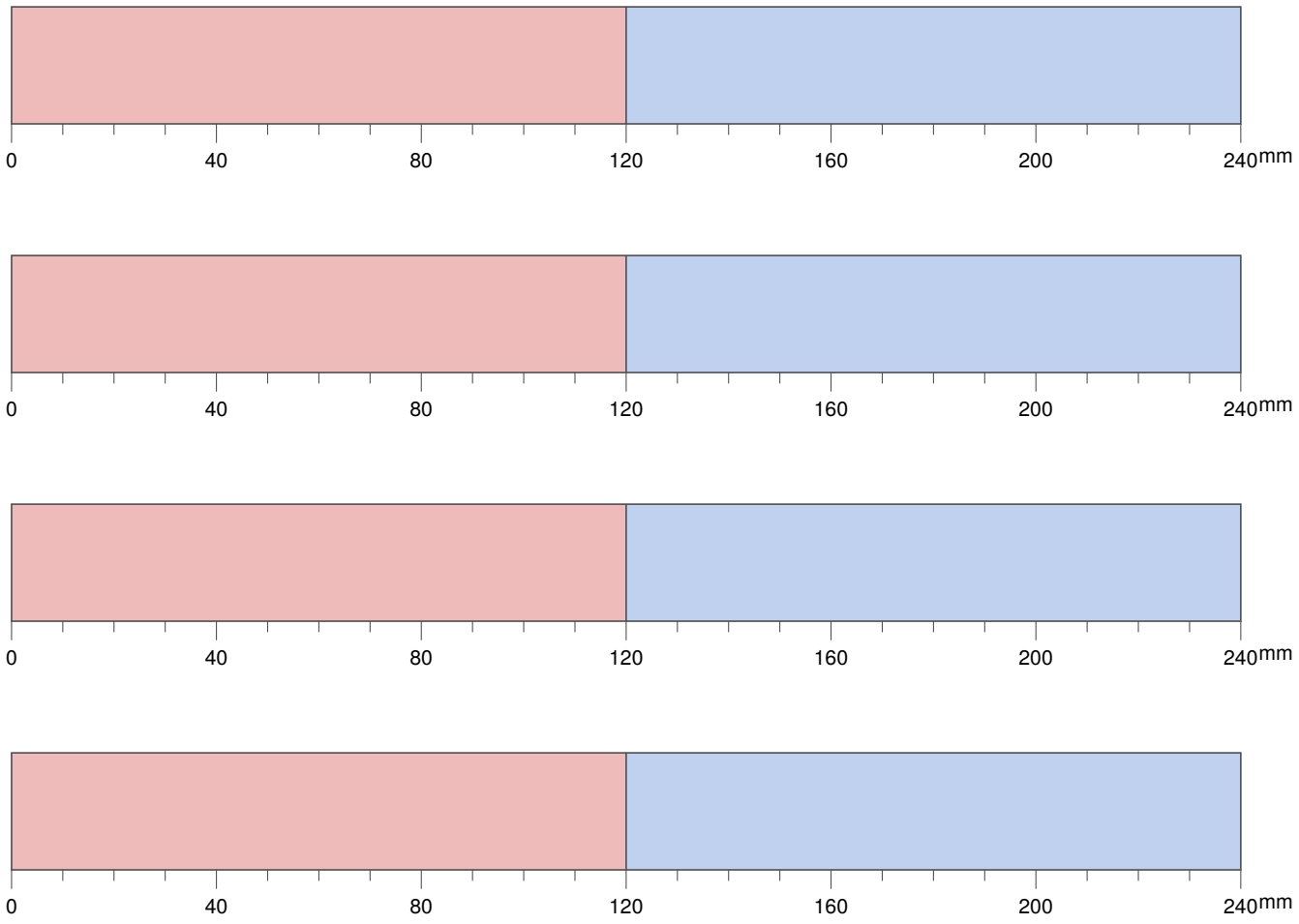
Problem 2: Share each of these trains with as few cuts as you can. If a train cannot be shared, cross it out.



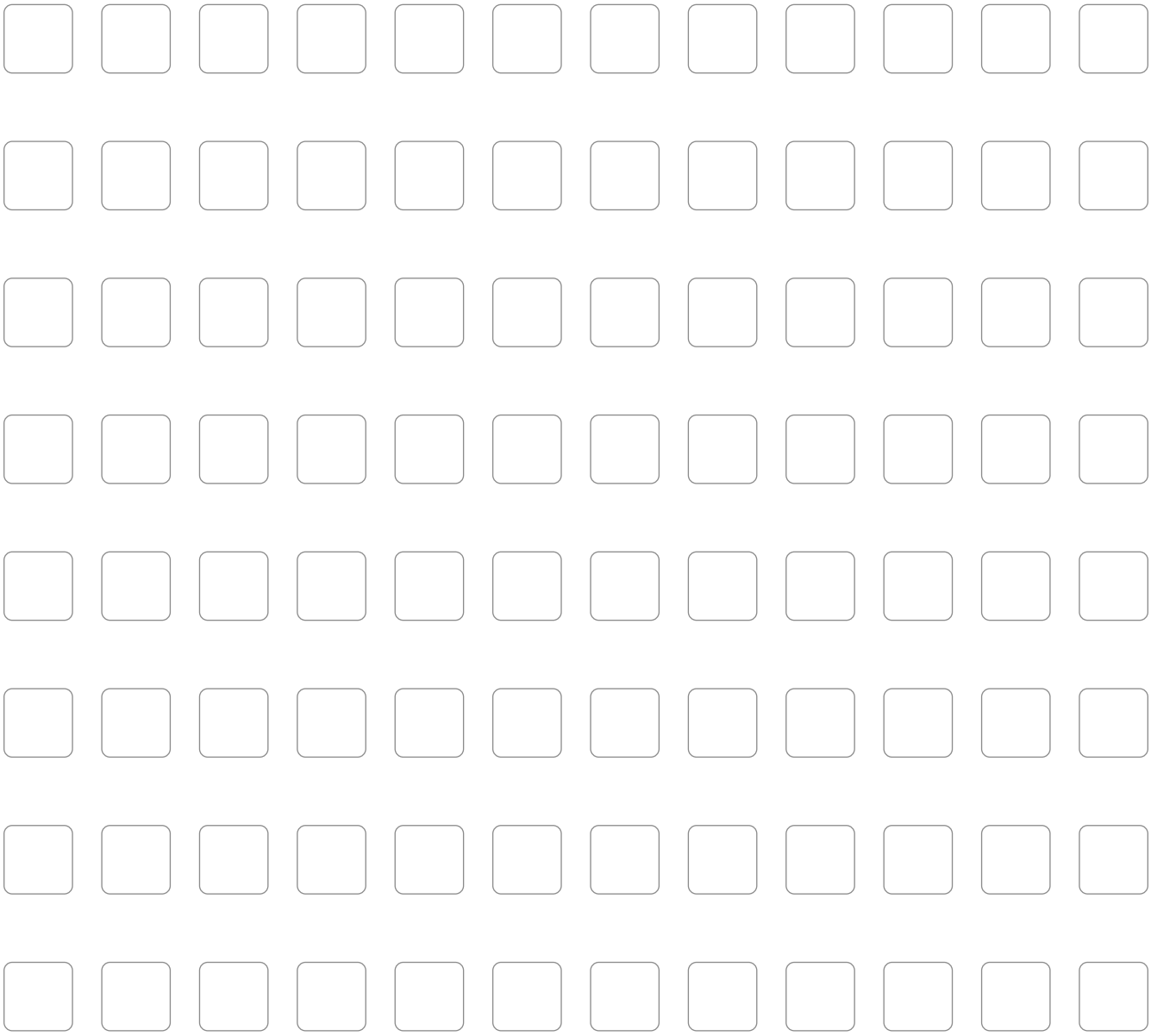
Problem 3: One person uses the two-person card A, and another person uses the two-person card B.



On a strip, each number is the value of the whole 120 mm panel of that colour, and value follows length: 40 mm of red is worth 1 to A. Every cut goes straight across the strip. Share one strip between two trays so that A finds the two trays worth the same, and B also finds them worth the same. Use as few cuts as you can. Can one cut do it? Measure on your strip from the red end, and mark in pencil before anyone cuts.



Problem 4: Can you make a train that needs three cuts to share it? Use an even number of red cards and an even number of blue cards.



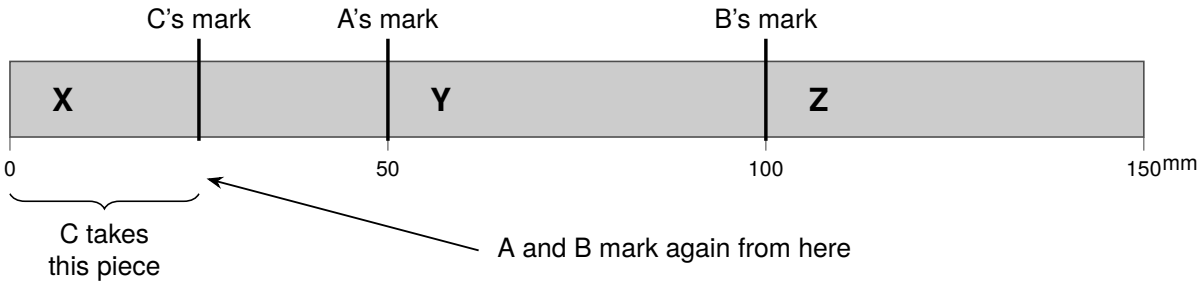
Problem 5: Three people use the three-person cards A, B and C, which give these values to the grey panels X, Y and Z.

	X	Y	Z
card A	4	8	0
card B	0	4	8
card C	8	0	4

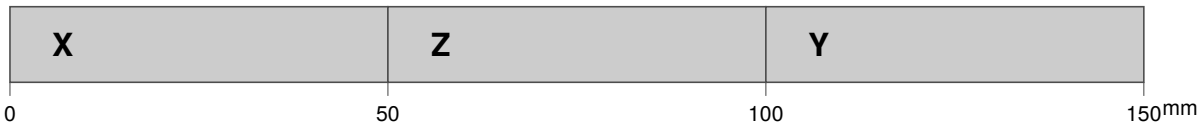
Each number is the value of a whole 50 mm panel. A panel may be cut straight across, and value follows length within a panel:



The three people share a row of the panels with a marking game, run by a referee. Each person marks where the piece from the left end first becomes worth 4 by their own card. Whoever marked leftmost takes that piece. The other two mark again from the new left end, aiming at 4 again, and whoever marked leftmost takes that piece. The last person takes the rest.



Play the game with the panels in the order X Z Y.



Problem 6: Play the game again with the panels in the orders Y X Z and Z Y X. In your three games, does anyone get less than 4 by their own card? Would anyone rather have someone else's piece?



